

How To Become A Cybersecurity Enginner



By SHAILESH SHAKYA @BEGINNERSBLOG.ORG



Phase 1: Building a Solid Foundation

1.1 Deep Dive into Core Concepts

Computer Science:

- **Programming:**

- **Python:** This is non-negotiable in cybersecurity. You'll use it for everything from automating security tasks to analyzing malware.
 - **Coursera:** "Python for Everybody Specialization" by the University of Michigan (Excellent for beginners, covers fundamental programming concepts and applies them to real-world problems, including data analysis relevant to security).
 - **Udemy:** "Complete Python Bootcamp From Zero to Hero in Python" by Jose Portilla (Very popular and comprehensive, with a focus on practical coding skills).
 - **Hands-on:** Practice by writing scripts to automate tasks like:
 - Scanning a network for open ports.
 - Parsing security logs to extract key information.
 - Implementing basic encryption and decryption algorithms.
- **Go:** Gaining traction in security due to its speed and efficiency, especially in areas like cloud-native security and DevOps.
 - **Udacity:** "Learn Go" (This Nanodegree program provides a structured learning path with hands-on projects).
 - **Hands-on:** Explore Go libraries for security and networking, and try building simple security tools.
- **JavaScript:** Essential for understanding web application security and browser-based attacks.
 - **edX:** "CS50's Web Programming with Python and JavaScript" by Harvard University (This course covers both Python and JavaScript, giving you a versatile skillset for web development and security).
 - **Hands-on:** Analyze JavaScript code in websites to understand how it can be exploited, and practice secure coding technique

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Operating Systems:

- **Linux:**

- The dominant OS in server environments and security infrastructure. You need in-depth knowledge of Linux.
- **edX:** "Introduction to Linux" by the Linux Foundation (A great starting point to learn basic commands, file system navigation, and user management).
- **Coursera:** "Linux Server Management and Security Specialization" by the University of Colorado System (This specialization dives deeper into Linux administration and security, covering topics like server hardening and security auditing).
- **Hands-on:**
 - Set up a virtual machine with a security-focused Linux distribution like Kali Linux or Parrot OS.
 - Practice using command-line tools for system administration, network analysis, and security hardening.

- **Windows:**

- While Linux is crucial, don't neglect Windows security. It's essential for endpoint protection and enterprise environments.
- **Udemy:** "Windows Security & Hardening: The Practical Guide" (Focuses on securing Windows systems against common threats).
- **Hands-on:**
 - Learn about Windows security features like Group Policy, User Account Control, and Windows Defender.
 - Practice configuring security settings and auditing security events in a Windows environment.

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Networking:

- **TCP/IP In-Depth:**

- Go beyond the basics and understand how data flows across networks.
- **Coursera:** "Computer Networking: A Top-Down Approach Specialization" by the University of Colorado Boulder (Comprehensive coverage of networking concepts, from the physical layer to application layer protocols).
- **Hands-on:**
 - Use Wireshark to capture and analyze network traffic, identify different protocols, and understand how data packets are structured.
 - Set up a home network with routers, switches, and firewalls to gain practical experience.

- **Network Security Essentials:**

- **Firewalls:**
 - Learn about different types (packet filtering, stateful, application-level), rule creation, and best practices for deployment.
 - **Hands-on:**
 - Set up a virtual network and configure firewalls using pfSense, OPNsense, or a cloud-based firewall solution.
- **VPN and Tunneling:**
 - Understand how VPNs establish secure connections and explore various protocols (OpenVPN, IPsec, WireGuard).
 - **Hands-on:**
 - Set up your own VPN server and connect to it from different devices.
- **Wireless Security:**
 - Master Wi-Fi security protocols (WPA2/3), encryption methods (AES), and secure configuration practices.
 - **Hands-on:**
 - Configure a Wi-Fi access point with strong security settings and experiment with different encryption methods.

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1.2 Security Concepts and Best Practices

- **Threat Modeling:** A proactive approach to security where you identify potential threats and vulnerabilities in systems.
 - **Book:** "Threat Modeling: Designing for Security" by Adam Shostack (A classic guide to threat modeling methodologies).
 - **Online Resources:** OWASP Threat Dragon (A free, open-source threat modeling tool).
 - **Hands-on:** Conduct threat modeling exercises for different systems or applications.
- **Security Architecture:** Learn to design secure systems by incorporating security principles like least privilege, defense in depth, and secure defaults.
 - **SANS Institute:** SEC401: "Security Essentials Bootcamp Style" (Provides a comprehensive introduction to security architecture and design).
- **Risk Management:** Understand how to assess and manage security risks using frameworks like NIST SP 800-30.
 - **Coursera:** "Risk Management for Cybersecurity" by the University of Washington (Covers risk assessment methodologies and risk mitigation strategies).

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1.3 Essential Tools and Skills

- **Command-Line Mastery:** Become fluent in the Linux command line (Bash). This is crucial for navigating systems, managing files, and using security tools.
 - **Hands-on:** Practice by using the command line for everyday tasks, automating tasks with shell scripts, and exploring different command-line tools.
- **Virtualization:** Master tools like VirtualBox, VMware, or Hyper-V to create isolated environments for practice, testing, and experimentation.
 - Hands-on: Set up virtual machines with different operating systems and configurations, and experiment with security tools and techniques in a safe environment.
- **Basic Scripting:** Start automating tasks and building simple security tools using Python or Bash scripting.
 - **Hands-on:** Write scripts to automate tasks like:
 - Gathering system information.
 - Scanning for vulnerabilities.
 - Performing basic network analysis.

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Phase 2: Exploring Core Security Domains

In this phase, you'll delve deeper into the key areas of cybersecurity, gaining practical skills and knowledge to protect systems and data.

2.1 Network Security (Advanced Techniques and Tools)

- **Intrusion Detection and Prevention Systems (IDPS):** Understand the difference between intrusion detection (IDS) and intrusion prevention (IPS) systems, and learn about signature-based, anomaly-based, and behavior-based detection methods.
 - **Tools:** Snort: A widely used open-source IDS/IPS.
 - **Hands-on:** Install and configure Snort to monitor network traffic, write custom rules to detect specific attacks, and analyze alerts.
 - Suricata: A high-performance IDS/IPS engine.
 - Hands-on: Explore Suricata's advanced features like multi-threading and rule optimization.
 - Zeek (formerly Bro): A powerful network security monitor for advanced threat detection and analysis.
 - Hands-on: Use Zeek to analyze network traffic, identify anomalies, and generate comprehensive logs for security investigations.
- **Advanced Firewall Configuration:** Next-Generation Firewalls (NGFWs): Learn about the capabilities of NGFWs, including application control, intrusion prevention, and deep packet inspection.
 - Hands-on: Explore NGFW features using virtual appliances or cloud-based firewalls from vendors like Palo Alto Networks or Fortinet.
- **Firewall Management:** Learn about firewall rule management, best practices for rule creation, and techniques for optimizing firewall performance.

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- **Wireless Security:** Wi-Fi Security Protocols: Deep dive into WPA2/3 security protocols, including their strengths, weaknesses, and configuration options.
- **Wireless Penetration Testing:** Learn how to assess the security of Wi-Fi networks using tools like Aircrack-ng. (Important: Only perform penetration testing on networks you have permission to test.)
 - Hands-on: Set up a Wi-Fi network with different security configurations and test its security using penetration testing tools.
 - Explore wireless security features in enterprise-grade access points.

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2.2 Endpoint Security (Protecting Devices and Data)

- **Endpoint Detection and Response (EDR):** Understand how EDR solutions work, including their ability to monitor endpoint activity, detect malicious behavior, and respond to threats in real-time.
- **Tools:**
 - **CrowdStrike Falcon:** A leading EDR solution with advanced threat hunting and incident response capabilities.
 - **Hands-on:** Explore CrowdStrike Falcon's features through free trials or community editions.
 - **SentinelOne:** Another popular EDR platform known for its autonomous response capabilities.
 - **Hands-on:** Experiment with SentinelOne's features and evaluate its effectiveness in detecting and responding to threats.
 - **Microsoft Defender for Endpoint:** A comprehensive endpoint security solution integrated with the Windows operating system.
 - **Hands-on:** Learn how to configure and manage Microsoft Defender for Endpoint in a Windows environment.

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- **Anti-Malware Techniques:** Static and Dynamic Analysis: Learn how to analyze malware samples to understand their behavior and identify indicators of compromise (IOCs).
 - **Tools:** VirusTotal: An online service that analyzes files and URLs against multiple antivirus engines.
 - **Hybrid Analysis:** A malware analysis platform that provides detailed reports on malware behavior.
- **Sandboxing:** Explore how sandboxing techniques can be used to safely execute and analyze malware in an isolated environment.
 - **Tools:** Cuckoo Sandbox, Any.Run
- **Data Loss Prevention (DLP):** Understand the importance of DLP in protecting sensitive data and learn about different DLP techniques (content inspection, data tagging, encryption).
 - **Tools:** Explore DLP solutions from vendors like Symantec, McAfee, and Forcepoint.
 - **Hands-on:** Implement basic DLP policies to prevent sensitive data from leaving the organization.

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2.3 Cloud Security (Securing the Cloud Frontier)

- **Cloud Security Fundamentals:** Shared Responsibility Model: Understand the shared responsibility model in cloud security, where cloud providers are responsible for securing the underlying infrastructure, and users are responsible for securing their applications and data.
- **Cloud Security Architectures:** Learn about secure cloud design principles, including network segmentation, access control, data encryption, and security monitoring.
- **Coursera:** "Cloud Computing Security" by the University of Illinois at Urbana-Champaign (This course provides a good overview of cloud security concepts and best practices).
- **Cloud-Native Security Tools:** AWS Security Hub: A central platform for managing security and compliance across AWS accounts.
 - **Hands-on:** Explore AWS Security Hub and learn how to configure security checks, analyze findings, and automate remediation.
- **Azure Security Center:** A unified security management system for Azure resources.
 - **Hands-on:** Use Azure Security Center to monitor security posture, identify vulnerabilities, and implement security recommendations.
- **Google Cloud Security Command Center:** A security and data risk platform for Google Cloud.
 - **Hands-on:** Learn how to use Security Command Center to detect threats, manage vulnerabilities, and monitor compliance.

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- **Container Security:** Docker and Kubernetes Security: Understand how to secure containerized applications and environments, including image scanning, runtime security, and network policies.
 - **Udemy:** "Docker and Kubernetes: The Complete Guide" by Stephen Grider (This course covers both Docker and Kubernetes, including security best practices).
 - **Hands-on:** Implement security best practices for building and deploying containerized applications.
 - Use security tools like Clair or Anchore Engine to scan container images for vulnerabilities.

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2.4 Offensive Security (Ethical Hacking and Penetration Testing)

- Penetration Testing Methodologies: OWASP Testing Guide: Learn about the OWASP (Open Web Application Security Project) testing methodology for web application penetration testing.
- **NIST SP 800-115:** Understand the NIST (National Institute of Standards and Technology) guidelines for conducting penetration testing.
- **PTES (Penetration Testing Execution Standard):** A comprehensive framework for conducting penetration tests.
- **Essential Tools:** Metasploit: A powerful penetration testing framework for exploiting vulnerabilities.
 - **Hands-on:** Learn how to use Metasploit to conduct penetration tests, develop exploits, and automate tasks.
- **Burp Suite:** A comprehensive suite of tools for web application security testing.
 - **Hands-on:** Use Burp Suite to intercept and modify HTTP requests, scan for vulnerabilities, and exploit web applications.
- **Nmap:** A versatile network scanner for discovering hosts and services on a network.
 - **Hands-on:** Use Nmap to scan networks, identify open ports, and fingerprint operating systems.

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- **Vulnerability Assessment:** Automated Vulnerability Scanners: Learn how to use automated vulnerability scanners like Nessus, OpenVAS, and QualysGuard to identify vulnerabilities in systems and applications.
- **Manual Vulnerability Analysis:** Develop skills in manual vulnerability analysis to identify weaknesses that automated tools might miss.
- **Hands-on Practice:** Vulnerable VMs: Practice your penetration testing skills on vulnerable virtual machines like Metasploitable 2, OWASP Broken Web Applications, and DVWA (Damn Vulnerable Web Application).
- **Capture the Flag (CTF) Challenges:** Participate in CTF competitions to test your skills in a fun and competitive environment.

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2.5 Data Security (Protecting Sensitive Information)

Database Security:

- Access Control: Implement access control mechanisms to restrict access to sensitive data in databases.
- Encryption: Learn about database encryption techniques to protect data at rest and in transit.
- Auditing: Configure database auditing to track data access and modifications.
- Hands-on: Practice securing databases using security features provided by database management systems like MySQL, PostgreSQL, and Microsoft SQL Server.

Data Masking and Anonymization:

Techniques: Learn about data masking techniques (substitution, shuffling, perturbation) and anonymization methods (k-anonymity, l-diversity) to protect sensitive data while preserving its utility.

- Tools: Explore data masking and anonymization tools to implement these techniques in real-world scenarios.
- Data Breach Response: Incident Response Plan: Develop an incident response plan for data breaches, including steps for containment, eradication, recovery, and post-incident activity.
- Digital Forensics: Learn basic digital forensics techniques to investigate data breaches and collect evidence.

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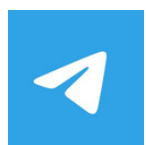
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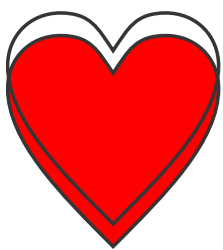
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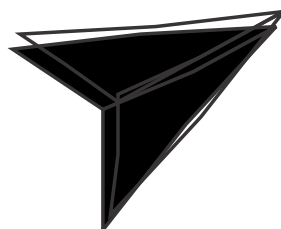
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